MILITARY SPECIFICATION

CASE SECTION, BALLISTIC, CENTER AFT: SECTIONS "C" AND "D"

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of case assembly used to house the Lance warhead section and is entitled Case Section, Ballistic, Center Aft: Sections "C" and "D".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

U.S. ARMY

8879856	- Case Assembly, "C" Section.
8879857	- Bond Assembly C Section.
8879858	- Ring, "C" Section, Station 32.875.
8879859	- Ring, Sta. 61.375.
8879870	- Case Assembly, "D" Section.
8879871	- Bond Assembly, D Section.
8879886	- Edge Member, Main Access Door.
8879901	- Ring, Sta. 100.00 "D" Section.
8879905	- Maintenance Access Door, Bond Assembly.
8879952	- Adhesive Bonding of Ballistic Case Sections "C" and "D".
8879960	- Cork, Bonding.
8879967	- Caulking, Conductive Compound and Sealer.
8879974	- Edge Member "C" Bond Assembly, Forward.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

8879975 - Edge Member "C" Bond Assembly, Aft.
8879976 - Edge Member "D" Bond Assembly, Forward.
8879977 - Edge Member "D" Bond Assembly, Aft.
9237212 - Simulated Warhead Section.
EL-8879856 - Equipment List.
EL-8879870 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The case sections shall comply with all requirements specified on the drawings (dwgs) 8879856 and 8879870, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Bonding. The bonded assemblies shall comply with all the requirements, procedures, and tests specified on drawing 8879952. The contractor shall furnish objective evidence (see 6.2) that he has complied fully with the above drawing (8879952) requirements. Bonded assemblies shall be inspected for quality of bonds when tested as specified in 4.4.1.

3.4.1 Acceptable bond voids. The following bond voids will be acceptable and not be considered defects:

3.4.1.1 Bond assembly, C section, dwg 8879857.

3.4.1.1.1 Splice plates, P/N 8879857-7 and P/N 8879857-8. Voids along the external edges of the skins to the splice plate 0.10 inch in width and 2 inches long. In addition, the total length of voids must not exceed 50 percent of the splice length.

3.4.1.1.2 Edgemember, "C" bond assembly, forward, dwg 8879974, to the skins. Voids along the external edges of the skin to the edgemember 0.10 inch wide without restriction to their total length.

3.4.1.1.3 Edgemember "C" bond assembly, aft, dwg 8879975, to the skins. Voids along the external edges of the skin to the edgemember 0.06 inch wide and 2 inches long. The total length of these voids may not exceed 2 inches per 4 inches of circumferential length of edge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

3.4.1.2 Bond assembly, D section, dwg 8879871.

3.4.1.2.1 Splice plate P/N 8879871-12, 8879871-13, 8879871-14 and 8879871-15. Voids along the external edges of the skins to the splice plate 0.06 inch in width and 2 inches long. However, the total length of these voids may not exceed 2 inches per any 4 inches of length of splice edge.

3.4.1.2.2 Edgemember "D" section, forward, dwg 8879976, to the skins. Voids along the external edges of the skin to the edgemember 0.06 inch in width and 2 inches long. However, the total length of these voids may not exceed 2 inches per 4 inches of circumferential length of edge.

3.4.1.2.3 Edgemember, "D" section, aft, dwg 8879977, to the skins. Voids along the external edges of the skin to the edgemember 0.06 inch in width and 1 inch long. However, the total length of these voids may not exceed 1 inch per 4 inches of circumferential length of edge.

3.4.1.2.4 Edgemember, maintenance access door, dwg 8879905 and Edgemember, Main Access Door dwg 8879886 to skins. Voids along the external edges of the skin to these edgemembers 0.06 inch in width and 1 inch long. The total length of these voids may not exceed 2 inches measured along the length of edge.

3.4.1.2.5 Skin to balance of edgemembers for other openings and inserts. Voids along external edges of the skin to edgemembers and inserts 0.06 inch in width and 2 inches long. The total length of these voids may not exceed 2 inches measured along the length of edge.

3.4.2 Bond void. Any bond void exceeding the allowable criteria specified in 3.4.1 shall be considered a defect and initial cause for rejection pending review by designated agency in the invitation for bids or request for proposals.

3.5 Environmental testing. The case sections shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The skin structures shall also comply with 3.4 before and after each environmental test.
Testing shall be as specified in 4.4.2. Tests shall be conducted in the order specified.

3.5.1 Vibration. The bonded assemblies, "C" & "D" Sections shall withstand the vibration conditions specified in Table 1, along the longitudinal axis only, when mounted and instrumented as part of, and in accordance with drawing 9237212. Following this test as specified in 4.4.2, the bonded assemblies shall be tested for compliance with 3.4 after being stabilized at ambient temperature. Data from this instrumentation shall be recorded and retained as permanent records of base data only. It will not be used as a basis for acceptance or rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

Table I

Resonant search vibration

Frequency (Hz)	Gravity Units (g's) Double Amplitude (da-in)	Time-Minutes And Sweep Rate
10-14	0.1" da ± .01"	At ambient temperature, 1 frequency sweep from 10-500-10 at 0.745 octave/min.
14-500	1.0 g's ± 0.1	

Sinusoidal vibration

Frequency (Hz)	Gravity Units (g's) Double Amplitude (da-in)	Time-Minutes And Sweep Rate
5-17	0.2" da ± 0.02"	Frequency sweep at 0.875 octave/min at ambient temperature, 3 frequency sweeps from 5-500-5 (15 min each sweep) for a total of 45 min. At -65°F and at 160°F one frequency sweep from 5-500-5 for a total of 15 min at -65°F and for 15 min at +160°F.
17-130	3.0 g's ± 0.3	
130-500	5.0 g's ± 0.5	

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise speci-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

fied in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwgs 8879856 and 8879870) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

Table II

First article inspection and tests

Inspection or Test
1. Classification of characteristics (see 4.3.2.1.1 and 4.3.2.1.3)
2. Bonding (see 3.4)
3. Vibration (see 3.5.1)
4. Bonding (see 3.4)
5. Classification of characteristics (see 4.3.2.1.2 and 4.3.2.1.4)

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one batch of products, one manufacturer of raw material or components, one production method or manufacturing operation. The entire lot shall be submitted as a unit for inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

4.3.2 Product inspection examination. 100 percent of all items produced shall be inspected for all major characteristics. One (1) sample of "C" and "D" section shall be selected at random from every five (5) units produced and inspected for all listed minor characteristics. Any characteristics found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Compartment assembly case section ballistic "C" (see dwg 8879856)* before assembly of joint seals, alignment pins and cork.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect smaller OD at Station 32.875	Gage
102.	Incorrect hole location for alignment pin Station 32.875	Gage
103.	Mislocation of angular hole in mounting ring Station 32.875 (8)	Gage
104.	Incorrect hole size in mounting ring Station 32.875 (8)	Gage
105.	Incorrect seal groove diameter	Gage
106.	Incorrect seal groove width	Gage
107.	Incorrect seal groove depth	Gage
108.	Mislocation of holes in mounting ring Station 61.375 (18)	Gage
109.	Incorrect hole size in mounting ring Station 61.375 (18)	Gage
110.	Mislocation of alignment pin holes Station 61.375 (2)	Gage
111.	Mislocation of angular holes in mounting ring Station 62.50 (20)	Gage
112.	Incorrect hole size of angular holes in mounting ring Station 62.50 (20)	Gage
113.	Incorrect hole diameter for alignment pin Station 62.50 (2)	Gage
114.	Incorrect hole location for alignment pin at Station 62.50 (2)	Gage
115.	Evidence of poor workmanship (see 3.6)	Visual
Minor:		
201.	Incorrect contour	Gage
202.	Incorrect application of sealant	Visual
203.	Incorrect application of interior finish	Visual

*Dimensions not shown on 8879856 are to be found on 8879858, 8879859, and 8879857.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

4.3.2.1.2 Compartment assembly case section ballistic "C" (see dwg 8879856).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect assembly height of alignment pin Station 32.875	Gage
102.	Incorrect concentricity at Station 33.375	Gage
103.	Incorrect assembly height of alignment pin Station 61.375	Gage
104.	Incorrect dimension at Station 62.00 (see note 17)	Gage
105.	Incorrect diameter of Station 33.375	Gage
106.	Improper bonding of cork (see 8879960)	Visual/Manual
107.	Incorrect contour	Gage
108.	Excessive waviness	Gage
109.	Evidence of poor workmanship (see 3.6)	Visual
110.	Incorrect weight	Gage
111.	Incorrect center of gravity (cg)	Gage
112.	Incorrect cork thickness	Gage
113.	Any component missing, damaged or improperly assembled	Visual
114.	Incorrect perpendicularity at Station 32.875	Gage
115.	Incorrect height of forward ring small outside diameter	Gage
116.	Incorrect concentricity of forward ring small outside diameter	Gage
117.	Incorrect length from Station 32.875	Gage
118.	Mislocation of detent in mounting ring (20 places)	Gage
Minor:		
201.	Improper application of exterior sealer	Visual
202.	Seals not cemented securely	Visual/Manual
203.	Identification marking missing, incorrect or illegible	Visual
204.	Exterior sealer damaged, exposing cork	Visual
205.	Incorrect inside diameter at Station 32.875	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

4.3.2.1.3 Compartment assembly case section ballistic "D" (see dwg 8879870) before assembly of pins, T-bolts, doors, covers, seals, and cork.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect mounting hole diameter Station 62.285 (20) (see 8879976)	Gage
102.	Incorrect forward seal groove width (see 8879976)	Gage
103.	Incorrect forward seal groove depth (see 8879976)	Gage
104.	Incorrect location of mounting holes for maintenance access door (26)	Gage
105.	Incorrect parallelism Station 85.00	Gage
106.	Incorrect location of ring Station 85.00	Gage
107.	Incorrect location of holes in flange Station 85.00 (14)	Gage
108.	Incorrect diameter of holes in flange Station 85.00 (14)	Gage
109.	Incorrect location of mounting holes "S" band antenna, (4 holes, 3 places)	Gage
110.	Incorrect location of alignment pin holes Station 85.00 (2 places)	Gage
111.	Incorrect angularity of T-bolt slots (4)	Gage
112.	Incorrect seal groove width aft "D" ring (2)	Gage
113.	Incorrect seal groove depth aft "D" ring (2) (see 8879901)	Gage
114.	Evidence of poor workmanship (see 3.6)	Visual
115.	Incorrect dimension from center of sealing surface to bolt latches for Quick Access Door (QAD)	Gage
116.	Incorrect dimension from center of sealing surface to Parameter Setting Door (PSD) latches for PSD	Gage
Minor:		
201.	Incorrect contour	Gage
202.	Incorrect application of interior finish	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

4.3.2.1.4 Compartment case section ballistic "D" (see dwg 8879870).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper height of alignment pins Station 62.50 (2)	Gage
102.	Incorrect location of alignment pins (2) Station 62.50	Gage
103.	Incorrect dimension at Station 63.0 (see note 22)	Gage
104.	Improper bonding of cork (see 8879960)	Visual/Manual
105.	Incorrect cork thickness	Gage
106.	Incorrect contour	Gage
107.	Excessive waviness	Gage
108.	Improper height of alignment pins Station 85.00 (2)	Gage
109.	Incorrect concentricity Station 99.50	Gage
110.	Incorrect outside diameter Station 100	Gage
111.	Improper location of T-bolt (4) slots (see 8879901)	Gage
112.	Excessive gaps (max)	Gage
113.	Incorrect steps (max)	Gage
114.	Improper mounting torque values	Gage
115.	Improper height of alignment pins (2) Station 100 (max)	Gage
116.	Improper height of shear pins (6) Station 100 (max)	Gage
117.	Incorrect weight	Gage
118.	Incorrect center of gravity (cg)	Gage
119.	Evidence of poor workmanship (see 3.6)	Visual
120.	Any component missing, damaged or improperly assembled	Visual/Manual
121.	Incorrect concentricity Station 62.285	Gage
122.	Incorrect parallelism Station 62.285	Gage
123.	Incorrect length of forward datum	Gage
124.	Incorrect location of mounting holes Station 62.285 (20) (see 8879976)	Gage
125.	Incorrect location of maintenance door edgemember (see 8879871)	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

126. Incorrect location of alignment pins (2) Station 100 (see 8879901).....Gage
127. Incorrect location of shear pins (6) Station 100 (see 8879901).....Gage
128. Incorrect size of connector hole Station 100 minimum (see 8879901).....Gage
129. Incorrect torques for opening and/ or securing Quick Access Door (QAD)....Gage
130. Incorrect torques for opening and/ or securing Parameter Setting Door (PSD).....Gage
131. Leakage of Quick Access Door.....Gage
132. Leakage of PSD Door.....Gage
133. Pressure balance holes missing (2) (see 8879902).....Visual
134. Improper torque on maintenance access door mounting bolts (26).....Gage
135. Improper filling of cut in edge-member at fore end of D Section (see dwg 8879871).....Visual

Minor:

201. Incorrect location of forward bonded ring joint (see 8879976).....Gage
202. Incorrect location of umbilical edgemember (see 8879971).....Gage
203. Incorrect location "S" band antenna edgemembers (3 pieces) (see 8879871).....Gage
204. Incorrect location of PSD door edgemember (see 8879871).....Gage
205. Improper installation of "S" band antenna plugs(3).....Visual
206. Improper installation of PSD door assembly.....Visual
207. Improper installation of quick access door.....Visual
208. Improper installation of maintenance access door.....Visual
209. Improper installation of umbilical adapter ring.....Visual
210. Improper location of T-bolt covers (4).....Visual
211. Improper installation of cover mounting plate.....Visual
212. Incorrect location of PAL edge-member (see 887990).....Gage
213. Incorrect location of access door edgemember (see 8879901).....Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

4.3.3 Testing. Each "C" bond assembly (P/N 8879857) and "D" case bond assembly (P/N 8879871) produced shall be subjected to the following test.

Test	Test Classification
Bonding	Major

4.3.3.1 If the selected "C" and "D" bonded assembly fails to comply with the applicable requirement, the lot represented by the sample in which failure occurred shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8879856, and 8879870).

When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Bonding. The quality of the bonds (voids) shall be determined by using suitable ultrasonic or equivalent measuring equipment.

4.4.2 Vibration. The "C" and "D" bonded assemblies shall be mounted as part of a dummy load (see dwg 9237212) and secured to a suitable fixture and vibrated at the levels specified in 3.5.1. Control of the "g" or "double amplitude" levels shall be held to the total complex vibration signal. No tracking filter in the control loop is required. The quality of bonds of bonded assemblies shall meet the post-test inspection requirement of 3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The case section shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The case section shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The case section shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2043A dated 6 March 1973, with Amendment 3, dated 21 April 1975.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1115-A067

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.